MSG No. 89.88 $P4221'$ [Type II, tetragonal]

Table 1: Wyckoff site: 1a, site symmetry: 4221'

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]

Table 2: Wyckoff site: 1b, site symmetry: 4221'

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]

Table 3: Wyckoff site: 1c, site symmetry: 4221'

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]

Table 4: Wyckoff site: 1d, site symmetry: 4221'

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]

Table 5: Wyckoff site: 2e, site symmetry: 222.1'

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	[1,4,5,6,9,12,13,14]
2	$\left[0,\ \frac{1}{2},\ 0\right]$	[2,3,7,8,10,11,15,16]

Table 6: Wyckoff site: 2f, site symmetry: 222.1'

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[1,4,5,6,9,12,13,14]
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	[2,3,7,8,10,11,15,16]

Table 7: Wyckoff site: 2g, site symmetry: 4..1'

No.	position	mapping
1	$[0,\ 0,\ z]$	[1,2,3,6,9,10,11,14]
2	$[0,\ 0,\ -z]$	[4,5,7,8,12,13,15,16]

Table 8: Wyckoff site: `2h`, site symmetry: `4..1'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1,2,3,6,9,10,11,14]`
2	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[4,5,7,8,12,13,15,16]`

Table 9: Wyckoff site: `4i`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	`[1,6,9,14]`
2	$\left[\frac{1}{2}, 0, z\right]$	`[2,3,10,11]`
3	$\left[0, \frac{1}{2}, -z\right]$	`[4,5,12,13]`
4	$\left[\frac{1}{2}, 0, -z\right]$	`[7,8,15,16]`

Table 10: Wyckoff site: `4j`, site symmetry: `..21'`

No.	position	mapping
1	$[x, x, 0]$	`[1,7,9,15]`
2	$[-x, x, 0]$	`[2,5,10,13]`
3	$[x, -x, 0]$	`[3,4,11,12]`
4	$[-x, -x, 0]$	`[6,8,14,16]`

Table 11: Wyckoff site: `4k`, site symmetry: `..21'`

No.	position	mapping
1	$\left[x, x, \frac{1}{2}\right]$	`[1,7,9,15]`
2	$\left[-x, x, \frac{1}{2}\right]$	`[2,5,10,13]`
3	$\left[x, -x, \frac{1}{2}\right]$	`[3,4,11,12]`
4	$\left[-x, -x, \frac{1}{2}\right]$	`[6,8,14,16]`

Table 12: Wyckoff site: `4l`, site symmetry: `.2.1'`

No.	position	mapping
1	$[x, 0, 0]$	`[1,4,9,12]`
2	$[0, x, 0]$	`[2,7,10,15]`
3	$[0, -x, 0]$	`[3,8,11,16]`
4	$[-x, 0, 0]$	`[5,6,13,14]`

Table 13: Wyckoff site: `4m`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[1,4,9,12]`
2	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	`[2,7,10,15]`
3	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	`[3,8,11,16]`

continued ...

Table 13

No.	position	mapping
4	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[5,6,13,14]`

Table 14: Wyckoff site: `4n`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,4,9,12]`
2	$\left[0,\ x,\ \frac{1}{2}\right]$	`[2,7,10,15]`
3	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[3,8,11,16]`
4	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[5,6,13,14]`

Table 15: Wyckoff site: `4o`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,4,9,12]`
2	$\left[\frac{1}{2},\ x,\ 0\right]$	`[2,7,10,15]`
3	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[3,8,11,16]`
4	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[5,6,13,14]`

Table 16: Wyckoff site: `8p`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,9]`
2	$[-y,\ x,\ z]$	`[2,10]`
3	$[y,\ -x,\ z]$	`[3,11]`
4	$[x,\ -y,\ -z]$	`[4,12]`
5	$[-x,\ y,\ -z]$	`[5,13]`
6	$[-x,\ -y,\ z]$	`[6,14]`
7	$[y,\ x,\ -z]$	`[7,15]`
8	$[-y,\ -x,\ -z]$	`[8,16]`